$$\int_{3\pi/4}^{\pi} \tan\theta \, d\theta$$

Areas between curves (6.1)

This chapter concerns applications of integration. The typical setup is as follows.

- We have some quantity (area, volume, etc.) we want to compute.
- We write this quantity as a definite integral.
- We use antidifferentiation to compute the definite integral (if possible).

Suppose we have two functions f, g defined on the interval $[a, b]$, and $f(x) \geq g(x)$ for all x on this interval. What is the area between the their graphs on this interval?

Theorem: If $f(x) \geq g(x)$ on for all $a \leq x \leq b$, then the area between the graphs of f and g on this interval is

$$\int_a^b (f(x) - g(x))\, dx$$

What is the area enclosed by the graphs of $y = x - 1$ and $y = 1 - x^2$?

What is the total area of the regions bounded by $y = \sin x$, $y = \cos x$, $x = 0$, $x = \pi$?

Sometimes, it is easier to integrate with respect to y.

Theorem: If $f(y) \geq g(y)$ on for all $a \leq y \leq b$, then the area between the graphs of f and g on this interval is

$$\int_a^b (f(y) - g(y))\, dy$$

What is the area enclosed by the graphs of $x = y^2 - 1$ and $x = 3 - y^2$?

What is the area enclosed by the curves $y = 2x$, $y = x$, and $xy = 1$?